

OS Lab

Signals

Program 3: SIGCHLD - Child Process Signal

```
lahari@DESKTOP-AIC33F7:~$ nano signal1.c
lahari@DESKTOP-AIC33F7:~$ gcc signal1.c -o signal1
lahari@DESKTOP-AIC33F7:~$ ./signal1
Parent is waiting...
Child is running...
Child is terminating...
SIGCHLD received: Child has terminated.
Parent finished.
lahari@DESKTOP-AIC33F7:~$ |
```

Program 4 - SIGPIPE - Broken Pipe

1. This one connects directly to the pipes you have already learned. SIGPIPE occurs when a process tries to write to a pipe after all readers have closed the read end.
2. SIGPIPE tells the writer that it is trying to write to a pipe whose read end has been closed.

```
lahari@DESKTOP-AIC33F7:~$ nano signal3.c
lahari@DESKTOP-AIC33F7:~$ gcc signal3.c -o signal3
lahari@DESKTOP-AIC33F7:~$ ./signal3
Writing to pipe...
lahari@DESKTOP-AIC33F7:~$ |
```

Program 5 - kill() program

Parent sends signal to child Program

```
lahari@DESKTOP-AIC33F7:~$ nano signal4.c
lahari@DESKTOP-AIC33F7:~$ gcc signal4.c -o signal4
lahari@DESKTOP-AIC33F7:~$ ./signal4
Child waiting...
Parent sending signal...
Child received SIGUSR1
lahari@DESKTOP-AIC33F7:~$ |
```

Program 6 - SIGABRT — Abort

```
Child received SIGSEGV
lahari@DESKTOP-AIC33F7:~$ nano signal5.c
lahari@DESKTOP-AIC33F7:~$ gcc signal5.c -o signal5
lahari@DESKTOP-AIC33F7:~$ ./signal5
Program started
Aborted (core dumped)
lahari@DESKTOP-AIC33F7:~$ |
```

Program 7: SIGSEGV - Invalid Memory Access

```
lahari@DESKTOP-AIC33F7:~$ nano signal6.c
lahari@DESKTOP-AIC33F7:~$ gcc signal6.c -o signal6
lahari@DESKTOP-AIC33F7:~$ ./signal6
Before error
Segmentation fault (core dumped)
lahari@DESKTOP-AIC33F7:~$ |
```